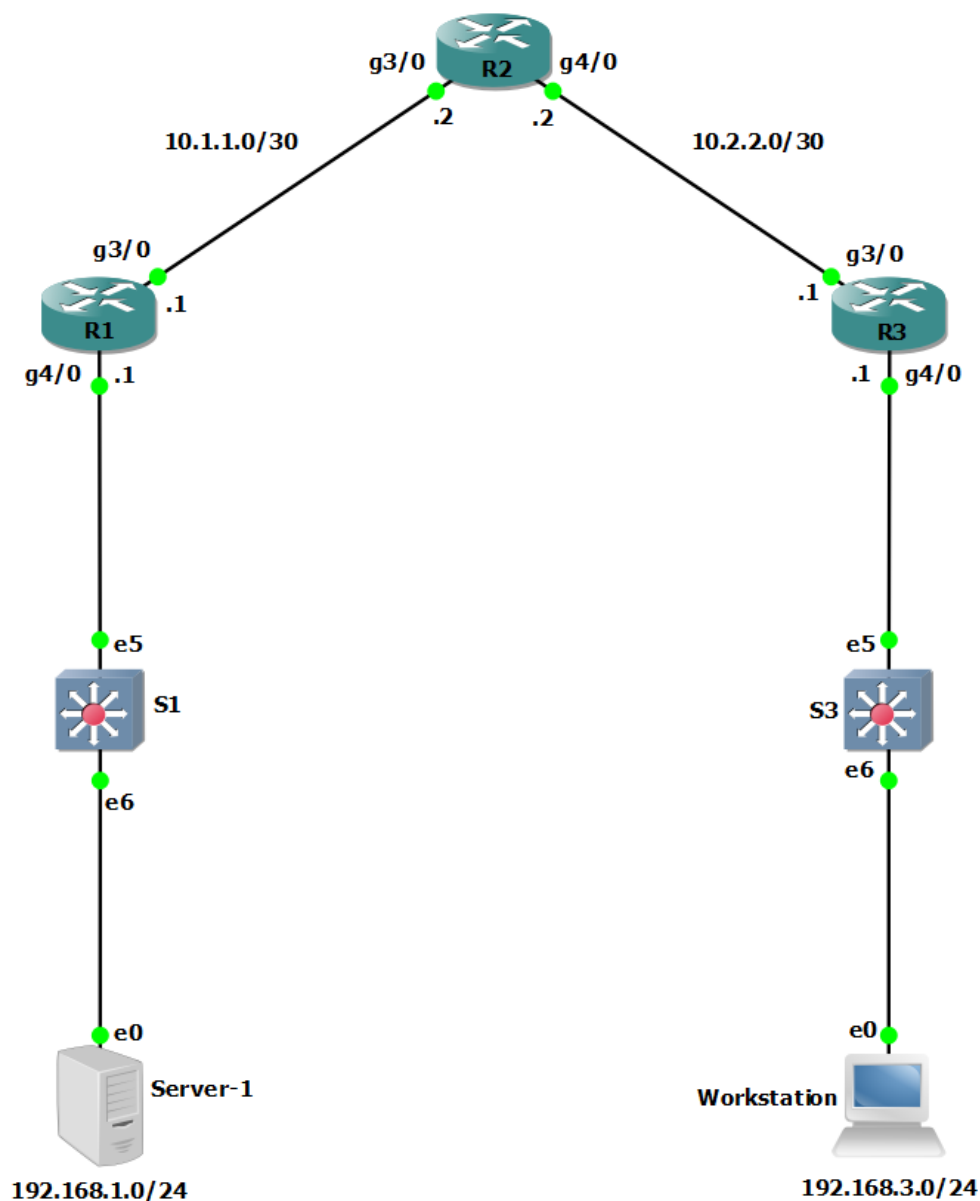


Lab - Configure Automated Security Features (DAYUTH THY)

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G3/0	10.1.1.1	255.255.255.252	N/A	N/A
	G4/0	192.168.1.1	255.255.255.0	N/A	S1 E5

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R2	G3/0	10.1.1.2	255.255.255.252	N/A	N/A
	G4/0	10.2.2.2	255.255.255.252	N/A	N/A
R3	G3/0	10.2.2.1	255.255.255.252	N/A	N/A
	G4/0	192.168.3.1	255.255.255.0	N/A	S3 E5
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 E6
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 E6

Objectives

Part 1: Configure Basic Device Settings

- Cable the network as shown in the topology.
- Configure basic IP addressing for routers and PCs.
- Configure OSPF routing.
- Configure PC hosts.
- Verify connectivity between hosts and routers.

Part 2: Configure Automated Security Features

- Lock down a router using AutoSecure and verify the configuration.
- Contrast using AutoSecure with manually securing a router using the command line.

Background / Scenario

The router is a critical component in any network. It controls the movement of data into and out of the network and between devices within the network. It is particularly important to protect network routers because the failure of a routing device could make sections of the network, or the entire network, inaccessible. Controlling access to routers and enabling reporting on routers is critical to network security and should be part of a comprehensive security policy.

In this lab, you will build a multi-router network and configure the routers and hosts. You will use automated security features on router R3.

Note: The routers used with hands-on labs are Cisco 4221 with Cisco IOS XE Release 16.9.6 (universalk9 image). The switches used in the labs are Cisco Catalyst 2960+ with Cisco IOS Release 15.2(7) (lanbasek9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Note: Before you begin, ensure that the routers and the switches have been erased and have no startup configurations.

Required Resources

- 3 Routers (Cisco 4221 with Cisco XE Release 16.9.6 universal image or comparable with a Security Technology Package license)
- 2 Switches (Cisco 2960+ with Cisco IOS Release 15.2(7) lanbasek9 image or comparable)
- 2 PCs (Windows OS with a terminal emulation program, such as PuTTY or Tera Term installed)

- Console cables to configure Cisco networking devices
- Ethernet cables as shown in the topology

Instructions

Part 1: Configure Basic Device Settings

In this part, set up the network topology and configure basic settings, such as interface IP addresses.

Step 1: Cable the network.

Attach the devices, as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for each router.

- a. Console into the router and enable privileged EXEC mode.

```
Router> enable
```

```
Router# configure terminal
```

Configure host names as shown in the topology.

```
R1(config)# hostname R1
```

Configure interface IP addresses as shown in the IP Addressing Table.

```
R1(config)# interface g0/0/0
```

```
R1(config-if)# ip address 10.1.1.1 255.255.255.0
```

```
R1(config-if)# no shutdown
```

```
R1(config)# interface g0/0/1
```

```
R1(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
R1(config-if)# no shutdown
```

```
interface GigabitEthernet3/0
ip address 10.1.1.1 255.255.255.252
negotiation auto
!
interface GigabitEthernet4/0
ip address 192.168.1.1 255.255.255.0
negotiation auto
!
interface GigabitEthernet5/0
no ip address
shutdown
negotiation auto
!
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 192.168.1.0 0.0.0.255 area 0
!
```

- b. To prevent the router from attempting to translate incorrectly entered commands as though they were host names, disable DNS lookup. R1 is shown here as an example.

```
R1(config)# no ip domain-lookup
no ip domain lookup
```

Step 3: Configure OSPF routing on the routers.

- a. Use the **router ospf** command in global configuration mode to enable OSPF on R1.

```
R1(config)# router ospf 1
```

- b. Configure the **network** statements for the networks on R1. Use an area ID of 0.

```
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
```

```
R1(config-router)# network 10.1.1.0 0.0.0.3 area 0
```

```
router ospf 1
passive-interface GigabitEthernet4/0
network 10.1.1.0 0.0.0.3 area 0
network 192.168.1.0 0.0.0.255 area 0
```

- c. Configure OSPF on R2 and R3.

```
R2(config)# router ospf 1
```

```
R2(config-router)# network 10.1.1.0 0.0.0.3 area 0
```

```
R2(config-router)# network 10.2.2.0 0.0.0.3 area 0
```

```
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 10.2.2.0 0.0.0.3 area 0
```

```
R3(config)# router ospf 1
```

```
R3(config-router)# network 10.2.2.0 0.0.0.3 area 0
```

```
R3(config-router)# network 192.168.3.0 0.0.0.255 area 0
```

```
router ospf 1
passive-interface GigabitEthernet4/0
network 10.2.2.0 0.0.0.3 area 0
network 192.168.3.0 0.0.0.255 area 0
```

- d. Issue the **passive-interface** command to change the G0/0/1 interface on R1 and R3 to passive.

```
R1(config)# router ospf 1
```

```
R1(config-router)# passive-interface g0/0/1
```

```
R1(config)#router ospf 1
R1(config-router)#pas
R1(config-router)#passive-interface g/4
^
% Invalid input detected at '^' marker.
R1(config-router)#passive-interface g4/0
```

```
R3(config)# router ospf 1
```

```
R3(config-router)# passive-interface g0/0/1
```

```
R3(config)#router ospf 1
R3(config-router)#pass
R3(config-router)#passive-interface g4/0
```

Step 4: Verify OSPF neighbors and routing information.

- Issue the **show ip ospf neighbor** command to verify that each router lists the other routers in the network as neighbors.

```
R1# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.2.2.2	1	FULL/BDR	00:00:37	10.1.1.2	GigabitEthernet0/0/0

```
R1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.2.2.2	1	FULL/BDR	00:00:39	10.1.1.2	GigabitEthernet3/0

- Issue the **show ip route** command to verify that all networks display in the routing table on all routers.

```
R1# show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```

10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.1.1.0/30 is directly connected, GigabitEthernet0/0/0
L       10.1.1.1/32 is directly connected, GigabitEthernet0/0/0
O       10.2.2.0/30 [110/2] via 10.1.1.2, 00:01:11, GigabitEthernet0/0/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0/1
•       192.168.3.0/24 [110/3] via 10.1.1.2, 00:01:07, GigabitEthernet0/0/0

```

```
R1#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

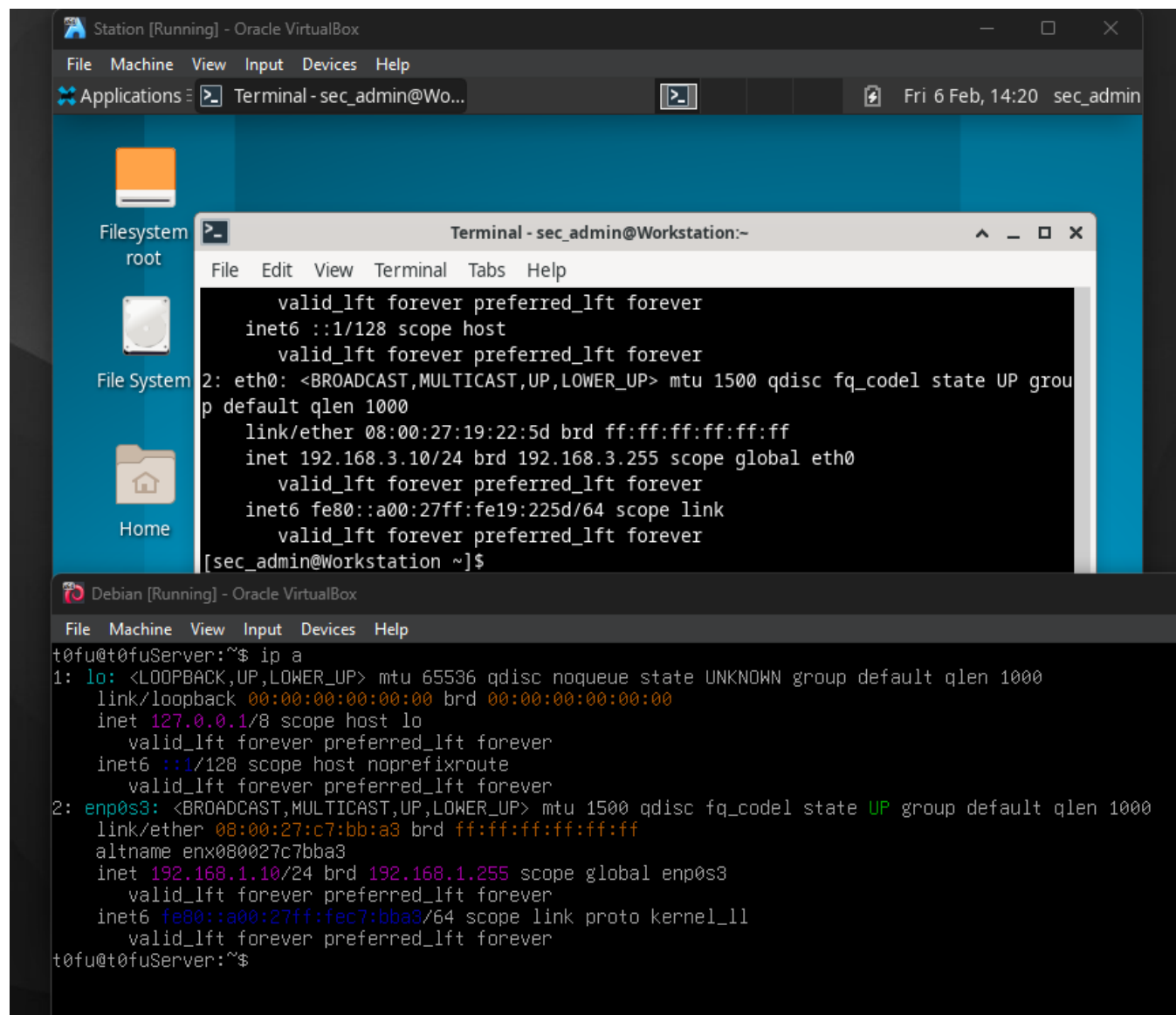
```

10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.1.1.0/30 is directly connected, GigabitEthernet3/0
L       10.1.1.1/32 is directly connected, GigabitEthernet3/0
O       10.2.2.0/30 [110/2] via 10.1.1.2, 01:56:14, GigabitEthernet3/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet4/0
L       192.168.1.1/32 is directly connected, GigabitEthernet4/0
O       192.168.3.0/24 [110/3] via 10.1.1.2, 01:43:25, GigabitEthernet3/0

```

Step 5: Configure PC host IP settings.

Configure a static IP address, subnet mask, and default gateway for PC-A and PC-C as shown in the IP Addressing Table.



Step 6: Verify connectivity between PC-A and PC-C.

- a. Ping from R1 to R3.

If the pings are not successful, troubleshoot the basic device configurations before continuing.

- b. Ping from PC-A, on the R1 LAN, to PC-C, on the R3 LAN.

If the pings are not successful, troubleshoot the basic device configurations before continuing.

Note: If you can ping from PC-A to PC-C you have demonstrated that OSPF routing is configured and functioning correctly. If you cannot ping but the device interfaces are up and IP addresses are correct, use the **show run**, **show ip ospf neighbor**, and **show ip route** commands to help identify routing protocol-related problems.

```
Debian [Running] - Oracle VirtualBox
File Machine View Input Devices Help
t0fu@t0fuServer:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c7:bb:a3 brd ff:ff:ff:ff:ff:ff
    altname enx080027c7bba3
    inet 192.168.1.10/24 brd 192.168.1.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fec7:bba3/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
t0fu@t0fuServer:~$ ping 10.1.1.1
PING 10.1.1.1 (10.1.1.1) 56(84) bytes of data.
64 bytes from 10.1.1.1: icmp_seq=1 ttl=255 time=9.02 ms
64 bytes from 10.1.1.1: icmp_seq=2 ttl=255 time=5.05 ms
64 bytes from 10.1.1.1: icmp_seq=3 ttl=255 time=9.74 ms
64 bytes from 10.1.1.1: icmp_seq=4 ttl=255 time=7.62 ms
64 bytes from 10.1.1.1: icmp_seq=5 ttl=255 time=4.03 ms
64 bytes from 10.1.1.1: icmp_seq=6 ttl=255 time=9.40 ms
^C
--- 10.1.1.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 500ms
rtt min/avg/max/mdev = 4.029/7.477/9.743/2.197 ms
t0fu@t0fuServer:~$ ping 192.168.3.1
PING 192.168.3.1 (192.168.3.1) 56(84) bytes of data.
64 bytes from 192.168.3.1: icmp_seq=1 ttl=253 time=46.8 ms
64 bytes from 192.168.3.1: icmp_seq=2 ttl=253 time=41.2 ms
64 bytes from 192.168.3.1: icmp_seq=3 ttl=253 time=38.5 ms
64 bytes from 192.168.3.1: icmp_seq=4 ttl=253 time=47.2 ms
^C
--- 192.168.3.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 300ms
rtt min/avg/max/mdev = 38.460/43.415/47.177/3.719 ms
t0fu@t0fuServer:~$ ping 192.168.3.10
PING 192.168.3.10 (192.168.3.10) 56(84) bytes of data.
64 bytes from 192.168.3.10: icmp_seq=1 ttl=61 time=45.9 ms
64 bytes from 192.168.3.10: icmp_seq=2 ttl=61 time=42.2 ms
64 bytes from 192.168.3.10: icmp_seq=3 ttl=61 time=61.0 ms
64 bytes from 192.168.3.10: icmp_seq=4 ttl=61 time=56.6 ms
64 bytes from 192.168.3.10: icmp_seq=5 ttl=61 time=56.1 ms
^C
--- 192.168.3.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 400ms
rtt min/avg/max/mdev = 42.237/52.389/61.038/7.094 ms
t0fu@t0fuServer:~$
```

Part 2: Configure Basic Security Settings on R1

In this part, copy and paste the following commands into R1 to configure basic security settings.

```
enable
configure terminal
service password-encryption
security passwords min-length 10
enable algorithm-type scrypt secret cisco12345
ip domain name netsec.com
```

```
username user01 algorithm-type scrypt secret user01pass
username admin privilege 15 algorithm-type scrypt secret adminpasswd
banner motd " Unauthorized access is strictly prohibited! "
line con 0
  exec-timeout 5 0
login local
  logging synchronous
line aux 0
  exec-timeout 5 0
login local
line vty 0 4
  exec-timeout 5 0
  privilege level 15
transport input ssh
  login local
crypto key generate rsa general-keys modulus 1024
ip ssh time-out 90
ip ssh authentication-retries 2
ip ssh version 2
```

Part 3: Configure Automated Security Features

In this part, you will do as follows:

- Use AutoSecure to secure R3.
- Review router security configurations with CLI.

By using a single command in CLI mode, the AutoSecure feature allows you to disable common IP services that can be exploited for network attacks. It can also enable IP services and features that can aid in the defense of a network when under attack. AutoSecure simplifies the security configuration of a router and hardens the router configuration.

Step 1: Use the AutoSecure Cisco IOS feature on R3.

- a. Enter privileged EXEC mode using the **enable** command.
- b. Issue the **auto secure** command on R3 to lock down the router. R2 represents an ISP router, so assume that R3 G0/0/0 is connected to the internet when prompted by the AutoSecure questions. Respond to the AutoSecure questions as shown in the following output. The responses are bolded.

```
R3# auto secure
--- AutoSecure Configuration ---

*** AutoSecure configuration enhances the security of
the router but it will not make router absolutely secure
from all security attacks ***
```

All the configuration done as part of AutoSecure will be shown here. For more details of why and how this configuration is useful, and any possible side effects, please refer to Cisco documentation of AutoSecure.

Lab - Securing the Router for Administrative Access

At any prompt you may enter '?' for help.
Use ctrl-c to abort this session at any prompt.

If this device is being managed by a network management station,
AutoSecure configuration may block network management traffic.
Continue with AutoSecure? [no]: **yes**

Gathering information about the router for AutoSecure

Is this router connected to internet? [no]: **yes**

Enter the number of interfaces facing internet [1]:

Interface	IP-Address	OK?	Method	Status
Protocol				
GigabitEthernet0/0/0	10.2.2.1	YES	manual	up
GigabitEthernet0/0/1	192.168.3.1	YES	manual	up
Serial0/1/0	unassigned	YES	unset	up
Serial0/1/1	unassigned	YES	unset	up

Enter the interface name that is facing internet: **GigabitEthernet0/0/0**

Securing Management plane services..

Disabling service finger
Disabling service pad
Disabling udp & tcp small servers
Enabling service password encryption
Enabling service tcp-keepalives-in
Enabling service tcp-keepalives-out
Disabling the cdp protocol

Disabling the bootp server
Disabling the http server
Disabling the finger service
Disabling source routing
Disabling gratuitous arp

Here is a sample Security Banner to be shown
at every access to device. Modify it to suit your
enterprise requirements.

Authorized Access only

This system is the property of So-&-So-Enterprise.
UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED.
You must have explicit permission to access this
device. All activities performed on this device
are logged. Any violations of access policy will result

in disciplinary action.

Enter the security banner {Put the banner between k and k, where k is any character}:

Unauthorized Access Prohibited

Enable secret is either not configured or is the same as the enable password

Enter the new enable secret: **cisco12345**

Confirm the enable secret : **cisco12345**

Enter the new enable password: **12345cisco**

Confirm the enable password: **12345cisco**

Configuration of local user database

Enter the username: **admin**

Enter the password: **adminpasswd**

Confirm the password: **adminpasswd**

Configuring AAA local authentication

Configuring console, Aux and vty lines for local authentication, exec-timeout, transport
Securing device against Login Attacks

Configure the following parameters

Blocking Period when Login Attack detected: **60**

Maximum Login failures with the device: **2**

Maximum time period for crossing the failed login attempts: **30**

Configure SSH server? [yes]: **[Enter]**

Enter the domain-name: **www.netsec.com**

Configuring interface specific AutoSecure services

Disabling the following ip services on all interfaces:

```
no ip redirects
no ip proxy-arp
no ip unreachable
no ip directed-broadcast
no ip mask-reply
```

Securing Forwarding plane services..

Enabling unicast rpf on all interfaces connected to internet

Configure CBAC Firewall feature? [yes/no]: **no**

This is the configuration generated:

```
no service finger
no service pad
no service udp-small-servers
no service tcp-small-servers
service password-encryption
service tcp-keepalives-in
service tcp-keepalives-out
no cdp run
no ip bootp server
no ip http server
no ip finger
no ip source-route
no ip gratuitous-arps
banner motd ^C  Unauthorized Access Prohibited ^C
security passwords min-length 6
security authentication failure rate 10 log
enable secret 5 $1$lubv$Rdx4gHUcijbxV7p2z76/71
enable password 7 110A1016141D5D5B5C737B
username admin password 7 02050D4808095E731F1A5C
aaa new-model
aaa authentication login local_auth local
line console 0
  login authentication local_auth
  exec-timeout 5 0
  transport output telnet
line aux 0
  login authentication local_auth
  exec-timeout 10 0
  transport output telnet
line vty 0 4
  login authentication local_auth
  transport input telnet
line tty 1
  login authentication local_auth
  exec-timeout 15 0
login block-for 60 attempts 2 within 30
ip domain-name www.netsec.com
crypto key generate rsa general-keys modulus 1024
ip ssh time-out 60
ip ssh authentication-retries 2
line vty 0 4
```

```
transport input ssh telnet
service timestamps debug datetime msec localtime show-timezone
service timestamps log datetime msec localtime show-timezone
logging facility local2
logging trap debugging
service sequence-numbers
logging console critical
logging buffered
int GigabitEthernet0/0/0
  no ip redirects
  no ip proxy-arp
  no ip unreachable
  no ip directed-broadcast
  no ip mask-reply
int GigabitEthernet0/0/1
  no ip redirects
  no ip proxy-arp
  no ip unreachable
  no ip directed-broadcast
  no ip mask-reply
ip access-list extended 100
  permit udp any any eq bootpc
interface GigabitEthernet0/0/0
  ip verify unicast source reachable-via rx 100
!
end
```

Apply this configuration to running-config? [yes]: **[Enter]**

Applying the config generated to running-config

WARNING: Command has been added to the configuration using a type 5 password. However, type 5 passwords will soon be deprecated. Migrate to a supported password type

WARNING: Command has been added to the configuration using a type 7 password. However, type 7 passwords will soon be deprecated. Migrate to a supported password type

WARNING: Command has been added to the configuration using a type 7 password. However, type 7 passwords will soon be deprecated. Migrate to a supported password typeThe name for the keys will be: R3.www.netsec.com

% The key modulus size is 1024 bits

% Generating 1024 bit RSA keys, keys will be non-exportable...

Lab - Securing the Router for Administrative Access

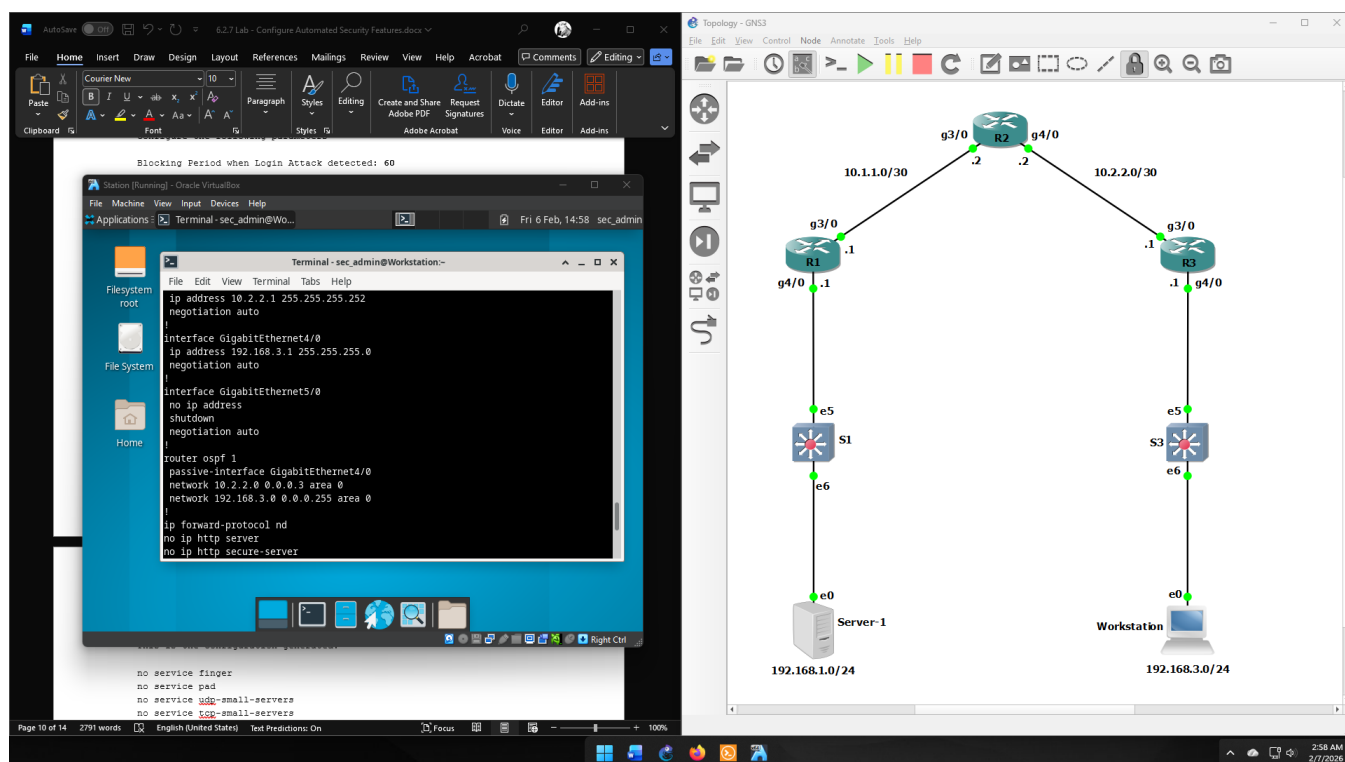
[OK] (elapsed time was 0 seconds)

R3#

Note: The questions asked and the output may vary depend on the features on the IOS image and device.

Step 2: Establish an SSH connection from PC-C to R3.

- Start PuTTY or another SSH client, and log in with the **admin** account and password **adminpasswd** created when AutoSecure was run. Enter the IP address of the R3 G0/0/1 interface **192.168.3.1**.
- Because SSH was configured using AutoSecure on R3, you will receive a PuTTY security warning. Click **Yes** to connect anyway.
- Enter privileged EXEC mode with password **cisco12345**, and verify the R3 configuration using the **show run** command.



Step 3: Contrast the AutoSecure-generated configuration of R3 with the manual configuration of R1.

- What security-related configuration changes were performed on R3 by AutoSecure that were not performed in previous sections of the lab on R1?

By running AutoSecure, R3 enabled AAA and created a named authentication list called local_auth to manage Console, AUX, and VTY logins. The router also hardened its security by disabling the HTTP server, CDP, and other exploitable services while enabling the TCP intercept feature and setting a minimum password length of six. Finally, it improved monitoring by configuring a failure rate log for authentication attempts and enabling critical logging for both the console and buffered memory.

2. What security-related configuration changes were performed in previous sections of the lab that were not performed by AutoSecure?

Answers will vary but could include: Telnet access was excluded from vty transport input. Additional accounts were created

3. Identify at least five unneeded services that were locked down by AutoSecure and at least three security measures applied to each interface.

Note: Some of the services listed as being disabled in the AutoSecure output above might not appear in the **show running-config** output because they are already disabled by default for this router and Cisco IOS version.

Services disabled include:

- no service pad
- no cdp run
- no ip bootp server
- no ip http server
- no ip source-route
- no ip gratuitous-arps
- no ip identd

For each interface, the following were disabled:

- no ip redirects
- no ip proxy-arp
- no ip unreachable

4. What are some advantages to using AutoSecure?

AutoSecure speeds up router hardening, reduces configuration errors, identifies overlooked vulnerabilities, and simplifies complex Cisco IOS security configurations.

Router Interface Summary Table

Router Model	Ethernet Interface #1	Ethernet Interface #2	Serial Interface #1	Serial Interface #2
1900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
4221	Gigabit Ethernet 0/0/0 (G0/0/0)	Gigabit Ethernet 0/0/1 (G0/0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)

Lab - Securing the Router for Administrative Access

Router Model	Ethernet Interface #1	Ethernet Interface #2	Serial Interface #1	Serial Interface #2
4300	Gigabit Ethernet 0/0/0 (G0/0/0)	Gigabit Ethernet 0/0/1 (G0/0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)

Note: To find out how the router is configured, look at the interfaces to identify the type of router and how many interfaces the router has. There is no way to effectively list all the combinations of configurations for each router class. This table includes identifiers for the possible combinations of Ethernet and Serial interfaces in the device. The table does not include any other type of interface, even though a specific router may contain one. An example of this might be an ISDN BRI interface. The string in parenthesis is the legal abbreviation that can be used in Cisco IOS commands to represent the interface.